AXI-Lite FIR DSP Peripheral

Modular VHDL IP core implementing a register-controlled
FIR DSP interface with scalable architecture.

Design specification

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1. Project definition

This project focuses on the design and implementation of a synthesizable AXI-Lite slave peripheral in VHDL intended to control an external FIR DSP module through a structured register interface and dedicated control signals. The objective is to model a realistic system-on-chip integration scenario in which a processor communicates with a digital signal processing block via a standard bus protocol, while the DSP itself is implemented as a separate RTL component. In the first phase of the project, data and configuration transfers to the DSP are handled through internal signals driven by memory-mapped registers; in a second phase, the architecture will be extended to support high-throughput data paths using DMA or AXI-Stream interfaces.

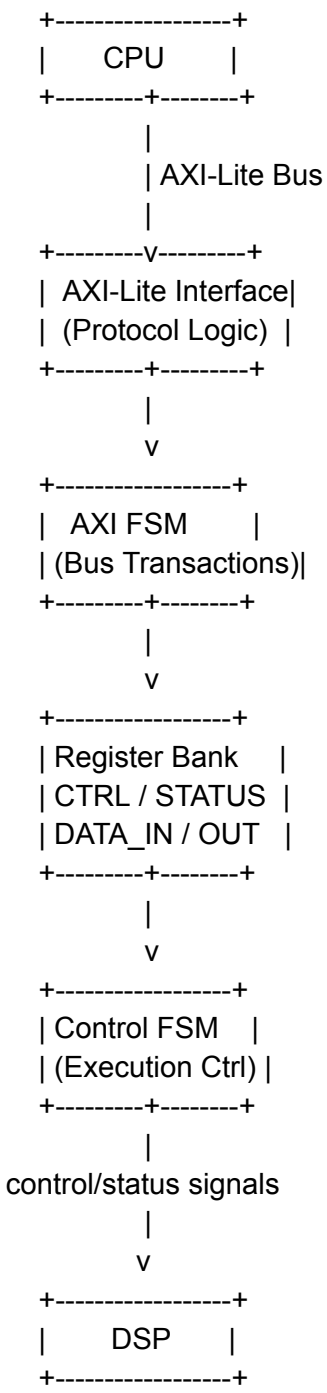
The peripheral exposes a set of memory-mapped registers accessible through AXI-Lite transactions, including control, status, coefficient, and configuration registers. A processor writes commands and parameters into these registers, and the peripheral's control logic interprets them to coordinate execution of the external FIR DSP. When enabled, the peripheral asserts control signals toward the DSP and monitors its status signals, such as ready or overflow, to supervise operation and ensure correct sequencing.

Internally, the design is partitioned into three main subsystems: an AXI protocol interface, a register bank, and a control finite state machine. The AXI interface handles address, data, and response channels using the VALID/READY handshake mechanism defined by the protocol. The register bank stores configuration values, coefficients, and control bits written by software, while the control FSM sequences interaction with the DSP, guaranteeing proper timing, synchronization, and safe coefficient updates. This separation of concerns improves readability, simplifies debugging, and allows each block to be verified independently.

The external FIR DSP module is connected to the peripheral through a dedicated signal interface that carries control, status, coefficient, and sample signals. Because the DSP resides outside the AXI peripheral module, it can be replaced, upgraded, or verified independently without modifying the bus interface. This modularity is essential in scalable hardware systems, where signal-processing cores may evolve or be swapped while maintaining a stable processor-facing control interface.

2. High-level overview

Block diagram



Module Architecture Description

TOP
├── axi_lite_slave_if
├── register_bank
├── control_fsm
├── dsp_if_adapter
└── dsp_pkg

Top-Level Module

The top-level module defines the external interface of the peripheral and structurally integrates all internal submodules. It instantiates the AXI-Lite slave interface, register bank, control finite state machine, DSP interface adapter, and shared package definitions, interconnecting them through clearly defined signals. This module contains no behavioral logic and serves strictly as a structural hierarchy layer to improve readability, maintainability, and system integration.

AXI-Lite Slave Interface

The AXI-Lite slave interface implements the full protocol required for communication with the processor, including independent read and write channels and VALID/READY handshake signaling. It decodes incoming transactions and converts them into simplified internal control signals such as read enable, write enable, address, and data buses. By isolating bus protocol handling in a dedicated block, the rest of the system remains independent of timing and signaling details specific to the AXI standard.

Register Bank

The register bank implements all memory-mapped registers visible to software, including control, status, coefficient, and configuration registers. It updates register values upon write transactions and returns stored values during read operations. The module also exposes individual register fields as structured outputs for internal logic, allowing other components to react directly to configuration changes. Its functionality is limited to storage and mapping, ensuring a clean separation from both protocol and execution logic.

Control Finite State Machine

The control FSM governs the operational sequencing of the external FIR DSP module. It interprets register contents, manages execution states, coordinates coefficient loading, and supervises DSP status signals such as ready or overflow. Based on these conditions, it generates deterministic control outputs including enable, reset, coefficient write strobes, and interrupt signals. This module encapsulates the behavioral logic of the peripheral and ensures safe, synchronized operation across all execution phases.

DSP Interface Adapter

The DSP interface adapter provides a dedicated interface layer between the internal control logic and the external DSP core. It formats signals, aligns handshakes, registers outputs, and performs any necessary timing adaptation to satisfy the DSP interface requirements.

This abstraction layer prevents tight coupling between system control logic and DSP implementation details, allowing the external processing module to be replaced or modified without impacting the rest of the design.

Shared Types and Constants Package

The shared package defines global constants, type declarations, register address maps, bit-field masks, and state enumerations used throughout the design. Centralizing these definitions ensures consistency across modules, reduces duplication, and simplifies maintenance. It also improves code clarity by replacing literal values with meaningful symbolic names, which is essential for scalable and reusable RTL development.

3. Requirements

Functional Requirements

FR-1 — The peripheral shall implement a fully compliant AXI-Lite slave interface supporting independent read and write channels.

FR-2 — The peripheral shall expose memory-mapped registers for control, status, configuration, and coefficient loading.

FR-3 — The peripheral shall allow software to enable and disable the DSP core.

FR-4 — The peripheral shall support runtime coefficient updates through register writes.

FR-5 — The peripheral shall provide real-time status feedback including ready, busy, and overflow indicators.

FR-6 — The peripheral shall prevent unsafe coefficient updates while DSP processing is active.

Interface Requirements

IR-1 — The module shall use a 32-bit AXI-Lite data bus.

IR-2 — The module shall support aligned 32-bit register accesses.

IR-3 — The DSP interface shall expose control, status, coefficient, and data signals.

IR-4 — All external signals shall be synchronous to the system clock.

Timing Requirements

TR-1 — All logic shall be synchronous to a single clock domain.

TR-2 — The design shall not include combinational paths between AXI inputs and outputs.

TR-3 — Register writes shall take effect within one clock cycle.

TR-4 — DSP control signals shall meet setup/hold requirements relative to the clock.

Reset Requirements

RR-1 — The design shall support an active-low synchronous reset.

RR-2 — After reset, all registers shall return to default values.

RR-3 — The DSP shall remain disabled after reset until explicitly enabled.

Safety and Robustness Requirements

SR-1 — Reserved register bits shall ignore writes and read as zero.

SR-2 — Invalid register accesses shall not cause undefined behavior.

SR-3 — Control signals shall never glitch during state transitions.

SR-4 — The system shall remain in a safe state if the DSP reports an error.

Scalability Requirements

SC-1 — The architecture shall allow replacement of the DSP module without modifying the AXI interface.

SC-2 — The design shall allow future addition of DMA or streaming data paths.

SC-3 — Register map shall reserve unused address space for future extensions.

4. Interface Specification

Inputs/outputs Ports

General

- **Clock and reset**
  clk : in std_logic
  rst_n : in std_logic

AXI-Lite Slave Interface

- **Write address**

  s_axi_awaddr : in std_logic_vector(31 downto 0)
  s_axi_awvalid : in std_logic
  s_axi_awready : out std_logic

- **Write data**

  s_axi_wdata : in std_logic_vector(31 downto 0)
  s_axi_wstrb : in std_logic_vector(3 downto 0)
  s_axi_wvalid : in std_logic
  s_axi_wready : out std_logic

- **Write response**

  s_axi_bresp : out std_logic_vector(1 downto 0)
  s_axi_bvalid : out std_logic
  s_axi_bready : in std_logic

- **Read address**

  s_axi_araddr : in std_logic_vector(31 downto 0)
  s_axi_arvalid : in std_logic
  s_axi_arready : out std_logic

- **Read data**

  s_axi_rdata : out std_logic_vector(31 downto 0)
  s_axi_rresp : out std_logic_vector(1 downto 0)
  s_axi_rvalid : out std_logic
  s_axi_rready : in std_logic

External DSP interface

- **Control**

  dsp_enable : out std_logic
  dsp_reset : out std_logic

dsp_mode : out std_logic_vector(1 downto 0)

- **Streaming data**

  dsp_data_in : out std_logic_vector(15 downto 0)
  dsp_data_in_valid : out std_logic
  dsp_data_in_ready : in std_logic

  dsp_data_out : in std_logic_vector(15 downto 0)
  dsp_data_out_valid : in std_logic
  dsp_data_out_ready : out std_logic

- **Coefficients**

  dsp_coeff_data : out std_logic_vector(15 downto 0)
  dsp_coeff_addr : out std_logic_vector(7 downto 0)
  dsp_coeff_we : out std_logic

Protocols

AXI-Lite Slave Protocol

The peripheral implements a fully compliant AXI4-Lite slave interface as defined by the AMBA AXI4 specification.

The interface supports independent read and write address/data channels using the VALID/READY handshake mechanism. All channels are synchronous to a single clock domain.

The following AXI-Lite features are supported:

- 32-bit data bus width
- 32-bit address width
- Single-beat transactions only
- Separate read and write channels
- Byte strobes (WSTRB) supported

The following features are not supported:

- Burst transactions
- Exclusive accesses
- Out-of-order responses
- Multiple outstanding transactions

All interface outputs are registered. No combinational paths exist between AXI inputs and outputs.

DSP Control Interface Protocol

The peripheral communicates with the external FIR DSP module using a simple synchronous control protocol.

The control protocol operates as follows:

1. Software writes configuration registers.
2. When the START condition is detected, the control FSM asserts dsp_start.
3. The DSP asserts dsp_busy while processing.
4. When computation completes, the DSP asserts dsp_done.
5. The FSM captures completion and updates status flags.

All signals are synchronous to the system clock. No asynchronous handshaking is used.

Coefficient updates are only permitted when dsp_busy = 0.

5. Functional Description

AXI-Lite Handshake

The interface supports independent read and write transactions using five separate channels:

- Write Address Channel (AW)
- Write Data Channel (W)
- Write Response Channel (B)
- Read Address Channel (AR)
- Read Data Channel (R)

All channels operate synchronously to a single clock domain and use the VALID/READY handshake mechanism.

Write Address Channel (AW)

The master asserts:

- *AWVALID* to indicate a valid address
- *AWADDR* containing the target register address

The slave asserts:

- *AWREADY* when it is ready to accept the address

The address transfer occurs when:

***AWVALID* = 1 and *AWREADY* = 1**

This condition defines a successful handshake. The slave registers the address on the rising edge of the clock when the handshake occurs.

Write Data Channel (W)

The master asserts:

- *WVALID*
- *WDATA*
- *WSTRB*

The slave asserts:

- *WREADY*

The data transfer occurs when:

***WVALID* = 1 and *WREADY* = 1**

Data is captured synchronously at the clock edge where the handshake occurs.

The slave supports byte-enable writes through the WSTRB signal. Only the selected byte lanes are updated during a write transaction, ensuring full AXI-Lite compliance.

Write Response Channel (B)

After both address and data have been accepted, the slave generates:

- *BVALID*
- *BRESP* (*OKAY* or *SLVERR*)

The master asserts:

- *BREADY*

The response transfer occurs when:

***BVALID* = 1 and *BREADY* = 1**

The slave deasserts *BVALID* after the handshake.

Read Address Channel (AR)

The master asserts:

- *ARVALID*
- *ARADDR*

The slave asserts:

- *ARREADY*

The address transfer occurs when:

***ARVALID* = 1 and *ARREADY* = 1**

Read Data Channel

After accepting the address, the slave returns:

- *RVALID*
- *RDATA*
- *RRESP*

The master asserts:

- *RREADY*

The data transfer completes when:

***RVALID* = 1 and *RREADY* = 1**

The slave deasserts RVALID after the handshake.

Handshake Rules

The *VALID*/*READY* protocol follows these rules:

1. *VALID* must remain asserted until *READY* is asserted.
2. *READY* may be asserted independently of *VALID*.
3. A transfer occurs only when both *VALID* and *READY* are high.
4. All outputs are registered and synchronous.
5. No combinational dependency exists between interface inputs and outputs.

Timing Behavior

- The slave supports single outstanding transactions.
- Only one write and one read transaction may be active at a time.
- Back-pressure is supported through *READY* deassertion.
- All address decoding is performed after the address handshake.

Error Handling

The slave returns:

- *OKAY* for valid aligned accesses
- *SLVERR* for invalid or misaligned addresses

DSP FSM

In the initial implementation phase, input samples are written by software through a memory-mapped DATA_IN register. Each write operation generates an internal strobe that feeds the DSP input interface using a ready/valid handshake mechanism. Output samples are captured into a DATA_OUT register for software retrieval. This approach simplifies integration while avoiding the complexity of streaming or DMA-based data movement.

READ/WRITE CONTROL

To send the input samples to the DSP Filter:

1. The AXI-Lite module would generate a sample_write_strobe flag when there is a new value in the DATA_IN register.

2. In the control DSP, if sample_write_strobe = ‘1’ and the DSP is enabled (CTRL.ENABLE = ‘1’), then dsp_data_in_valid is asserted and the sample is sent to dsp_data_in.
3. If the DSP asserts the dsp_data_in_ready then the dsp_data_in_valid is set to ‘0’.

To read the result from the DSP filter:

1. If dsp_data_out_valid is ‘1’ then the dsp_data_out is saved into the DATA_OUT register. The dsp_data_out_ready is set to 1.
2. In the next clock cycle the dsp_data_out_ready is reset.

To update the coefficients:

1. The AXI-lite module would generate a coef_write_strobe when there is a new value in the COEFF_DATA. If in this moment the STATUS.BUSY = ‘0’ and CTRL.ENABLE = ‘0’: the COEFF_DATA value is sent to dsp_coeff_data and the COEFF_ADDR value to dsp_coeff_addr. dsp_coeff_we is asserted for one cycle, as well as STATUS.BUSY.

The FSM states would be:

- IDLE
- SEND
- WAIT_RESULT
- LOAD_COEFF

STATUS.BUSY would be asserted if the FSM state is different from IDLE and the contrary on STATUS.READY.

This method is inherently slow, fully CPU-dependent, and does not scale for high-throughput applications. Since each sample transfer requires explicit software intervention through memory-mapped register writes and reads, overall performance is limited by processor bandwidth and latency. However, despite these limitations, this approach is well suited for the phase 1 of this project, where simplicity, observability, and ease of debugging are prioritized over performance. It enables straightforward validation of the control logic, DSP integration, and register interface before introducing more complex data movement mechanisms such as DMA or AXI-Stream in later development stages.

6. Register Map

Address	Name	Bits	Access	Reset	Description
0x00	CTRL	[0]	RW	0	Enable DSP
		[1]	RW	0	Reset DSP
		[3:2]	RW	0	Mode select 00: Normal FIR filtering 01: Bypass 10: Test pattern 11: Reserved
		[31:4]	—	0	Reserved
0x04	STATUS	[0]	RO	0	Ready
		[1]	RO	0	Error
		[2]	RO	0	Busy
		[31:3]	—	0	Reserved
0x08	COEFF_DATA	[15:0]	RW	0	Coefficient value
		[31:16]	—	0	Reserved
0x0C	COEFF_ADDR	[7:0]	RW	0	Coefficient index
		[31:8]	—	0	Reserved
0x10	DATA_IN	[31:0]	RW	0	DSP Data in
0x14	DATA_OUT	[31:0]	RO	0	DSP Data out

7. Reset and Initialization Behavior

The design implements a synchronous, active-low reset signal (rst_n). All internal state elements are reset on the rising edge of the system clock when rst_n is asserted low. The reset is assumed to be stable and synchronous to the clock domain.

Register Reset Values

After a reset:

- The DSP is disabled: CTRL.ENABLE = ‘0’
- All configuration registers are cleared to default values.
- The peripheral reports READY = ‘1’.
- Internal flags signals are cleared: sample_write_strobe and coef_write_strobe.
- No pending AXI transactions remain active.

AXI Interface Behavior During Reset

During reset assertion:

- All AXI output signals are driven to zero.
- The slave does not acknowledge transactions.
- Any ongoing transaction is aborted and must be reissued by the master after reset deassertion.

Reset Release Sequencing

After reset deassertion:

- The control FSM transitions to the IDLE state.
- The peripheral remains inactive until explicitly enabled by the CPU.
- No automatic coefficient loading is performed.
- The CTRL.ENABLE keeps inactive until the CPU changes it.

Verification Strategy

a. testbench approach
b. coverage plan
c. assertions
d. corner cases

Future Extensions

e. DMA support
f. streaming mode
g. interrupts